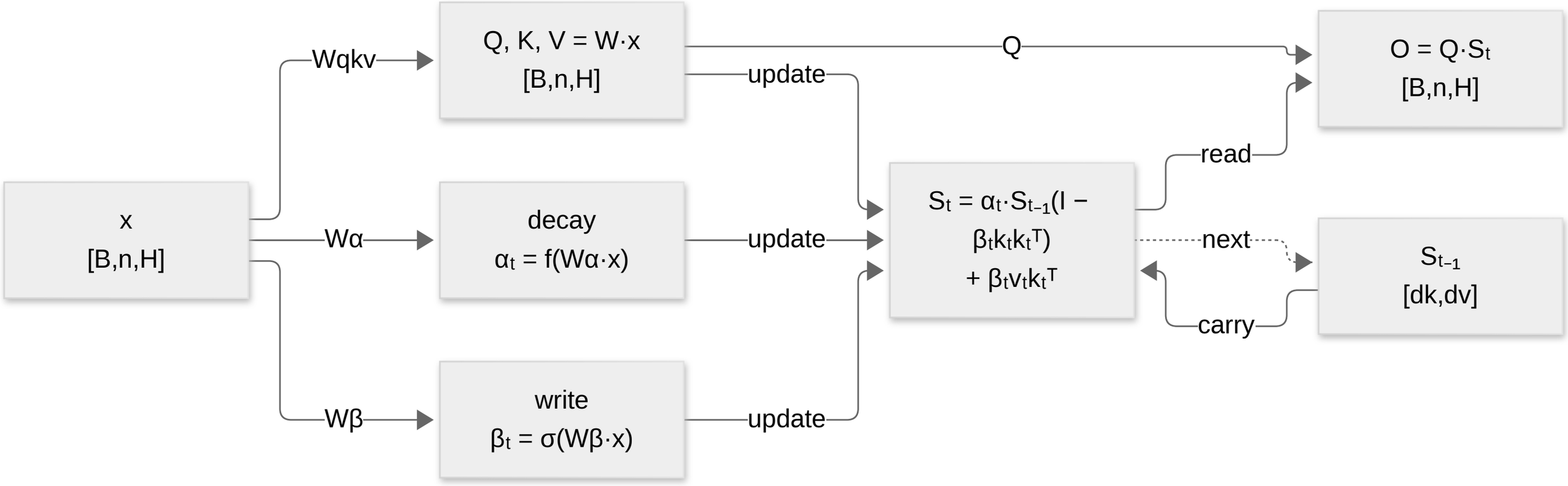

x
[B,n,H]
Wqkv
Wα
Wβ
Q, K, V = W·x
[B,n,H]
decay
$\alpha_t = f(W\alpha \cdot x)$
write
$\beta_t = \sigma(W\beta \cdot x)$
update
update
update
Q
$S_t = \alpha_t \cdot S_{t-1}(I - \beta_t k_t k_t^T) + \beta_t v_t k_t^T$
read
next
carry
$O = Q \cdot S_t$
[B,n,H]
S_{t-1}
[dk,dv]